2. Idea of water as a moderator: **(0.2 pts)**

3. Idea of self-regulating behavior: **(0.5 pts)**

4. Observation that power doubles when flow doubles: **(0.5 pts)**

iv) *(2 points)* The amount of energy generated by the reactor was νE_0. We shall consider the vast majority of it to have gone into heating up and vaporizing water. Let us consider water flowing in at $0°C$ and leaving at $100°C$. Then, the amount of energy absorbed by 1 kg of water is $(100°K)c + L = E_w = 2.68 \times 10^6$ J. The total mass of water that flowed into Oklo's reactor is $\nu E_0 / E_w = 9.09 \times 10^{13}$ kg.

Grading: (preliminary)

1. Approximation that all energy went into water **(0.5 p)**

2. $\Delta T \approx 100$ **(0.3 p)**

3. Both heating and vaporization considered **(0.3 p)**

4. $E_w = 2.68 \times 10^6$ J or similar idea **(0.3 p)**

5. $\nu E_0 / E_w = 9.09 \times 10^{13}$ kg **(0.6 p)**